

Resilient Cyber-Security Approach For Aviation Cyber-Physical Systems Protection Against Sensor Spoofing Attacks

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Abstract— The aviation industries are transitioning from conventional aircraft systems to Aviation Cyber-Physical Systems (ACPS) based aircraft. However, like any Cyber-Physical Systems (CPS), the ACPS are vulnerable to cyber-attacks that can be mounted by adversaries through the communication network infrastructure. This paper proposes a novel and resilient security protocol for detecting and defending ACPS against sensor spoofing cyber-attacks. First, a communication environment was developed to establish an aircraft Networked Control System (NCS) using the SimEvents toolbox. Then, a cyber-attack detection algorithm based on the positive selection of the Artificial Immune System (AIS) approach was developed and used to detect and drop suspicious communication packets on the aircraft network traffic. Finally, the NCS and the detection algorithm were integrated and tested on real cyber-security attack scenarios. The algorithm's accuracy was 0.96 based on the True Positive and True Negative algorithm detection rate. For further defending the aircraft against cyber-attacks, Nonlinear Autoregressive Exogenous (NARX) algorithm was developed to reconstruct or estimate the network dropped packets. The estimation accuracy of the NARX reached 0.99 using the coefficient of determination (R-value) based on the linear regression approach. The real-time simulation test results showed that the sensor spoofing cyber-attack was successfully detected. Also, the communication network of the ACPS was defended against the attack because the ACPS was maintaining the normal performance during the course of the cyber-attack.

Keywords— Sensor Spoofing Attacks, CPS, NARX, ACPS, Artificial Immune System, Positive Selection Algorithm, TCM, Cyber-attacks, FDI Attacks

I. INTRODUCTION

Aviation Cyber-Physical Systems (ACPS) provides more connectivity for several aircraft domains, such as aircraft control, airline information services, passenger information, and entertainment services [1]. Therefore, transforming traditional aircraft into ACPS increases aircraft connectivity and capability [2]. However, the integration of cyber components into the aircraft leads to possible unauthorized access to the aircraft's physical

components such as sensors, actuators, and control systems, hence increasing their vulnerability to cyber-attacks [3]. Generally, any CPS's security risks will be increased due to the integration of the communication system because cyber-attacks can be committed against the connected physical devices of ACPS. Cyber-attacks are significant threats to the Confidentiality, Integrity, and Availability (CIA) triad of the aircraft's physical devices [4]. Cyber-attacks cause tremendous threats to the operational stability and safety of ACPS. Therefore, more research investigation is needed to improve and enhance the cyber-security of ACPS due to the fact that the current cyber-security of physical systems have not reached maturity yet [5]. In particular, the cyber-security of ACPS is insufficient and requires further research and investigation. The security of ACPS is the top priority of the aviation industry [6]. It was reported that a team of US Department of Homeland Security (DHS) had successfully hacked a parked DHS-owned Boeing 757 without physical access or being on-board [7]. In 2008 a Spanair Flight 5022 had a crash due to a malware attack that impacted the on-board central computer [8]. This accident killed around 154 people. Also, the 2015 DoS attack on a Polish LOT airline flight was responsible for about 22 canceled and delayed flights at the Warsaw Chopin Airport [8]. The attacker targeted the computer system that accountable for sending critical flight plans to each aircraft before takeoff.

In this paper, we propose a novel and resilient security approach to detect and defend against sensor spoofing cyber-attacks launched on a sensor through the communication network of ACPS. The sensor spoofing attack was implemented in the form of a False Data Injection (FDI) attack. First, a communication system was developed to establish an NCS using SimEvents MATLAB/Simulink toolbox. This transformed the commercial aircraft testbed model into ACPS. Therefore, signals are exchanged between the physical system (aircraft model), sensors, actuators, and the control system in the form of communication network packets. The attack

detection algorithm was designed based on the AIS using the positive selection concept to detect sensor spoofing cyber-attacks. The AIS positive selection detection algorithm was added to the ACPS to distinguish between suspicious and normal network packets. As a result, suspicious network packets will be detected and dropped out of the communication network. Finally, a NARX neural network was developed and used as a predictor to estimate the dropped network packets by the detection algorithm.

The paper is organized as follows: In section II, a literature review is represented. In section III, the system architecture is discussed. In Section IV, we explained the proposed research methodology, and in Section V we discussed the system implementation, real-time simulation, and test results. Finally, in Section VI, we provide a conclusion of the research work.

II. LITERATURE REVIEW

This section reviews research works investigating the development of a communication network in the Cyber-Physical Systems (CPS) framework. It also discusses existing research concepts developed for detecting and defending against cyber-attacks in the CPS.

Because of the aircraft industry's interest in using Wireless Networked Control Systems (WNCSs), we will explain the basic architecture and operation of WNCS. The dynamic of WNCS was discussed in [9] is illustrated in Fig.1. The plant generates output signal $y(t)$, which is typically measured by the sensor. The time delay " τ^s " refers to the time that takes a sensor to transmit the network packet. Sensor readings are sent to the controller through a wireless network. In the controller, " τ^c " refers to the time needed to transmit the controller network packet and $r(t)$ refers to the reference signal. The controller sends the control signal " u " through the same wireless network. " τ^{ac} " refers to the delay in the communication system, and " α " refers to the switching function. Finally, the signal will reach the actuator, and " τ^a " refers to the time needed for the actuator to read the transmitted signal.

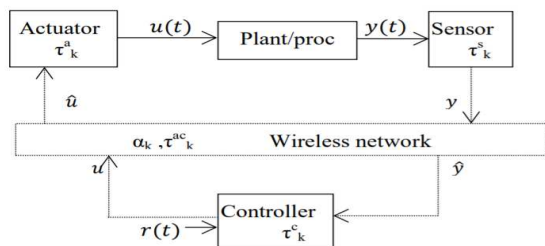


Fig. 1. Operation of the WNCS [9]

Authors in [10] discussed the impact of attacking sensors that are accountable for reading data used for Predictive Maintenance (PdM). They used PdM to analyze production data to forecast the health and the failure of a turbofan engine. This technique is also called the Remaining Useful Life (RUL) of the engine. In the experiments, the authors attacked

three sensors responsible for providing data to the PdM by modifying their readings. Three Deep learning algorithms were used to prevent the attack, which is: Gated Recurrent Unit (GRU), Convolutional Neural Network (CNN), and Long Short-Term Memory (LSTM). According to the authors, the GRU algorithm has the best performance among the used algorithms. Deep learning is a very efficient technique to classify and cluster data. However, it is still challenging to select an appropriate deep learning rule because no standard theory guides researchers [11].

Authors in the reference [12] discussed several sensors attack methods that can threaten sensors in Unmanned Aerial Vehicle (UAV) systems, which are GPS spoofing attacks and spoofing on other sensors attacks. GPS spoofing attacks consider as the most common attacks in UAV systems. An attacker can easily manipulate the GPS signals because they are unencrypted and unauthenticated signals used by the public. Spoofing on other sensors attacks includes attacking on-board sensors such as infrared, the barometer and gyroscope, radar, and electro-optical sensors. The main reason for this type of attack is to transfer the UAV system from a stable to an unstable system. An attacker can use ultrasonic waves to disturb the operation of the gyroscopic sensor. This forces the UAV system's attitude control to work insufficiently, which causing the UAV to crash.

III. SYSTEM ARCHITECTURE

In this section, we discuss the architecture of the ACPS that is used in this research.

A. Communication Network Simulator

The communication network structure follows the Client/Server topology, and it was designed based on the NCS approach. Fig. 2 shows the architecture of the NCS used in this research. The sensors, actuators, and pilot commands are considered the client-side, and the control system is considered the server-side. The communication system consists of three nodes: Node 1, Node 2, and Node 3. Sensors measure the aircraft's dynamics (state variables) " $y(k)$ " and transmit the measurements using Node1. Node 1 transmits " $y_c(k)$ ", which is the state variables signals of the aircraft in the form of network packets to the control system. Meanwhile, the control system receives the pilot commands " $r_c(k)$ " using Node 2. The control system computes and sends the control signals " $u(k)$ " to Node 3. Node 3 transmits " $u_c(k)$ ", which is the control signals packet to the actuators. Finally, the actuators transform the signals into mechanical movements.

The physical system used in the research is the Transport Class Model (TCM) of an aircraft. It is a commercial aircraft model developed by NASA for research purposes [13]. The control system used is the Model Reference Adaptive Control (MRAC) and was developed by the authors' previous research [14]. The communication system is simulated using the SimEvents library. SimEvents is a MATLAB/Simulink library that can model a communication system based on discrete-time events [15]. Three communication nodes are

developed using SimEvents to simulate a real-time network system based on a discrete-event approach. The communication architecture follows the Client/Server approach. The server has six ports to receive data from six clients and two ports to transmit data to the actuators, as shown in Tables I and II.

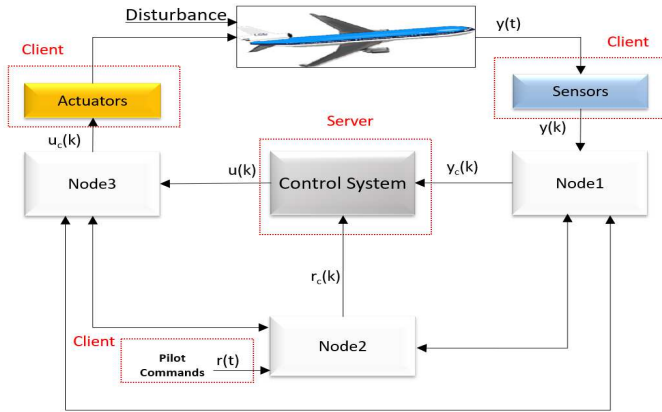


Fig. 2. The architecture of Networked Control System (NCS)

TABLE I. DATA RECEIVED BY THE SERVER

Port Number	Coming From	Data	Node Name
1	Sensor	State Variable "u"	Node 1
2	Sensor	State Variable "w"	Node 1
3	Sensor	State Variable "q"	Node 1
4	Sensor	State Variable "Θ"	Node 1
5	Pilot Command	Elevator signal	Node 2
6	Pilot Command	Throttle signal	Node 2

TABLE II. DATA SEND BY THE SERVER

Port Number	Going To	Data	Node Name
1	Actuator	Controller elevator signal	Node 3
2	Actuator	Controller Throttle signal	Node 3

1) Node 1

Node1 transmits "y(k)" network packet, which is the sensors readings of the longitudinal model of the aircraft, and it consists of the following state variables:

- The state variable "u" represents the velocity of the aircraft in the x-direction in ft/sec.
- The state variable "w" refers to the velocity of the aircraft in the z-direction in ft/sec.
- The state variable "q" represents the aircraft body pitch rate in rad/sec.
- The state variable "Θ" refers to the Euler angle for aircraft pitch in rad.

2) Node 2

It transmits the "r(k)" network packet, which contains the following pilot commands:

- Elevator command signal.
- Throttle command signal.

3) Node 3

It transmits "u(k)" network packet, which is the outputs signals of the control system:

- Controller elevator signal.
- Controller Throttle signal.

It was assumed that only Node 1 is compromised by the attackers to simulate the sensor spoofing attacks.

IV. RESEARCH METHODOLOGY

A. Cyber-Attack Detections Method

AIS is an intelligent machine learning algorithm that was design based on the theory of Natural Immune Systems (NIS) of the human body. The classical view of AIS supports both negative selection and positive selection concepts. The positive selection concept consists of the opposite approach to the negative selection concept. This research focuses only on the positive selection approach [16]. In the positive selection concept, the Artificial Lymphocytes (ALCs) algorithms are trained to classify self and non-self (antigen) cells [17].

1) Training of ALC Cells

Algorithm-1, shown in Fig.3. was used to train the ALC cells using a positive selection approach. The true-input (original sensor reading of the state variable "u") and the input (false sensor reading of the state variable "u") are initiated in step-1. The start and the end criteria of the training are set in steps 2-3. The size of the ALC is defined in step-4. In step-5, ALC is generated randomly, and in step-6, the threshold is set. Training the ALC cells begins from step-7 and ends at step-22. This loop aims to train the ALC to classify self ALC cells from non-self ALC cells (antigens).

2) Detection Procedure

Algorithm-2, shown in Fig. 4., was used to detect the spoofing attack using the trained ALC cells. The injected input was called in step-1. In step-2, the differ matrix is defined to store the calculated differences between the injected input and the trained ALC. The ALC_test matrix to store only the minimum values of the difference matrix was defined in step-3. The size of the ALC_test is set in step-4, and the trained ALC is called in step-5. The threshold is set in step-6. Steps 7-15 preforms the detection procedure to detect suspicious network packets using the specified threshold.

Algorithm 1: Training of ALC cells using Positive Selection

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1: Call true_input and injected_input
2: Set the counter n which is the start of the training
3: Set the counter en which is end of training ALC
4: Set the size of the ALC matrix row , column
5: Randomly generate ALC (row, col)
6: Set the threshold
7: while n <= en
8:   for i =1:row
9:     for j =1:col
10:      if ALC(i,j)>= true_input (i)+ threshold
11:        ALC(i,j)= random(1,1)
12:      else if ALC(i,j)<= true_input (i)- threshold
13:        ALC(i,j)= random(1,1)
14:      else
15:        ALC(i,j)= ALC(i,j)
16:      End if
17:    j++
18:  End for
19:  i++
20: End for
21: n=n+1
22: End while

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Fig. 3. The Training of ALC

Algorithm 2: Detecting of sensor attack

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1: injected_input //use the same input generated from training step
2: Set the differ=0 //to store the differences between the input and the ALC
3: Set the ALC_test=0 //to store only the smallest differences
4: Set the size of the ALC matrix row , column
5: ALC (row, col) //use the trained ALC from the training step
6: set the threshold
7: for i =1:row
8:   for j =1: col
9:     differ (i,j)=abs(injected_input (i,1)-ALC(i,j))
10:   End for
11:   ALC_test(i)=min(differ(i,end)) //only a small differences is matter
12:   if ALC_test(i)>= threshold
13:     malicious packet is detected and dropped
14:   End if
15: End for

```

Fig. 4. The Detection Algorithm

Fig. 5 illustrates the original sensor readings and the attacked (spoofed) sensor readings of the state variable “u” for 500 seconds. The sensor spoofing attack was injected between 200 and 400 seconds. Fig. 6 shows the results of the detection algorithm. The black “x” indicates the time when the attack was detected by the detection algorithm.

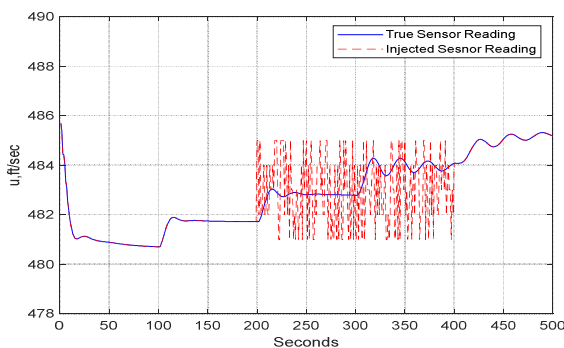


Fig. 5. Original and Injected Signals

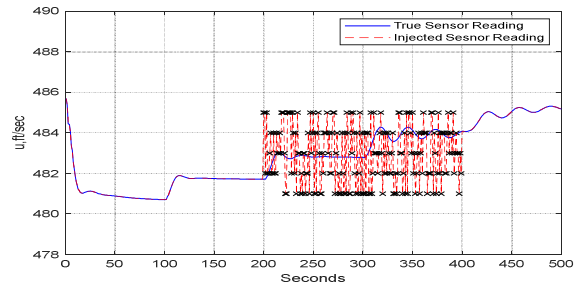


Fig. 6. Detection Result

3) Testing the Detection AIS Algorithm

To measure the performance accuracy of the AIS algorithm, true positive rate (TPR) and true negative rate (TNR) measures were used [18].

a) Detection True Positive Rate (TPR)

The equation used to calculate the TPR is shown in (1):

$$TPR = \frac{TP}{TP+FN} \quad (1)$$

TP is the true positive detection result, and FN is the false negative detection result. Table III. presents the number of TP and FN values. We run the algorithm five times, and then we took the average to provide better accuracy calculation. The total number of samples represents the total number of classified packets. It should be around 200 because the attack was injected at every second for 200 seconds, as was stated earlier.

TABLE III. DETECTION TPR

Number of Runs	Total of Samples	TP	FN	TPR
1	205	198	7	0.965
2	204	198	6	0.97
3	206	198	8	0.96
4	209	200	9	0.956
5	209	199	9	0.956
Average	206.6	198.6	7.8	0.96

b) Detection True Negative Rate (TNR)

The equation used to calculate the TNR is shown in (2)

$$TNR = \frac{TN}{TN+FP} \quad (2)$$

Where TN is the true negative detection result, and FP is the false positive detection result. Table IV presents the TN and FP result values. We run the algorithm five times, and then we took the average.

TABLE IV. DETECTION TNR

Number of Runs	Total of Samples	TN	FP	TNR
1	205	198	7	0.965
2	204	198	6	0.97
3	206	198	8	0.96
4	209	200	9	0.956
5	209	199	9	0.956
Average	206.6	198.6	7.8	0.96

The detection performance accuracy is calculated using the following equation:

$$ACC = \frac{TP+TN}{TP+TN+FP+FN} \quad (3)$$

Which produced an overall accuracy of the algorithm as:

$$ACC = (198.6+198.6)/(198.6+198.6+7.8+7.8) = 0.9622$$

B. Attack Defense Algorithm

1) NARX Neural Network

NARX algorithm was used to defend ACPS against the sensor spoofing attack. The NARX neural network was design based on series-parallel architecture. The developed NARX network comprises two nodes in the input layer, ten nodes in the hidden layer, and one node in the output layer [19]. The first input node receives the attacked (spoofed) sensor readings of the state variable “u” for 500. The second node receives the true reading of the sensor. We used the Levenberg-Marquardt (LM) algorithm to train the NARX neural network. LM training algorithm is used because it overcomes the issues of previous optimization algorithms such as Gradient descent and Gauss-Newton, such as a slow training process [20]. The LM algorithm merges the advantages of Gauss-Newton and gradient-descent methods to accomplish the optimal solution faster than both optimization methods [20]. LM is defined in (4).

$$W_{k+1} = W_k - [J^T J + \mu I]^{-1} J^T e \quad (4)$$

Where “ W_k ” is the network weight matrix, $J^T J$ is the approximation Hessian matrix, $J^T e$ represents the gradient descent, and “ J ” is the Jaconine matrix. “ e ” is a vector of network errors, and μ is a scalar. When μ is equal to 0, then (4) will only use the Hessian matrix term, which is reduced to Newton’s method; however, when μ is larger, the gradient descent method is used [21].

2) Testing the Defense Algorithm

As mentioned, the neural network was trained using the LM algorithm. A total of 50,000 input data points were used, 70% of the input data used for training, 15% for testing, and 15% for validation. Fig. 7 illustrates the regression plot of the training, testing, and validation. By looking at the figure, it can be observed that the coefficient of determination (R-value) for training, validation, and testing is higher than 0.99.

This means that the predicted output of the neural network is very close to the target data.

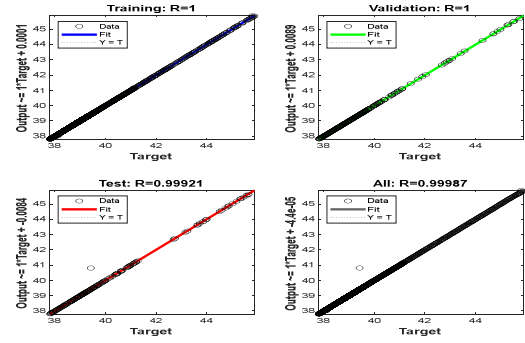


Fig. 7. NARX Training, Validation, Test, and overall performance Results

V. SIMULATION AND TEST RESULTS

The simulation process of the integrated ACPS system is represented in Fig. 8. We assumed that only Node 1 of the communication network is under attack, and the other nodes perform normally. The FDI attack was injected into the sensor responsible for reading the state variable “u” during the period from 200 to 400 seconds of the simulation. The state variable “u” represents the velocity of aircraft in the x-direction. After the attack was injected, the detection procedure is activated using the AIS algorithm to find and drop all suspicious packets. Then, the NARX neural network is used to compute estimates of the dropped packets.

In order to have an accurate simulation, initial conditions were given to the state variables of the longitudinal aircraft model, as shown in Table V. The initial altitude value was 10,000 ft, and the initial Calibrated Airspeed (cas) value of the aircraft was 250 knots.

TABLE V. AIRCRAFT LONGITUDINAL INITIAL CONDITIONS

State variables	Trim condition
u	485.69 ft/sec
w	38.72 ft/sec
q	0 rad/sec
Θ	0.079 rad

We present the effect of the cyber-attack on the ACPS and the validity of the proposed approach on the aircraft performance by conducting two simulation scenarios, Scenario-1 and Scenario-2, explained next.

In Scenario-1, we compared the response of the longitudinal aircraft model when the aircraft under attack with the reference model signals. The reference model provides information about where the aircraft should be. The aircraft angle of attack (AOA) was also compared when the aircraft is under sensor spoofing attack to the aircraft reference model. The aircraft performances in Scenario-1 are shown in Fig.9, Fig 10, Fig.11, Fig.12, and Fig.13. From the simulation test results, we can clearly conclude that the performance of the aircraft was unstable and unacceptable,

especially between the period of 200 and 400 seconds because of the attack.

In Scenario-2, we compared the responses of the longitudinal aircraft model under the sensor spoofing attack using the proposed approach with the reference model. We also compared the AOA performance in both cases. Fig.14, Fig.15, Fig.16, Fig.17, and Fig.18 illustrate the aircraft performances of Scenario-2. From the simulation test results, we can clearly observe that the proposed approach successfully maintained the aircraft’s stability and performed acceptable performance.

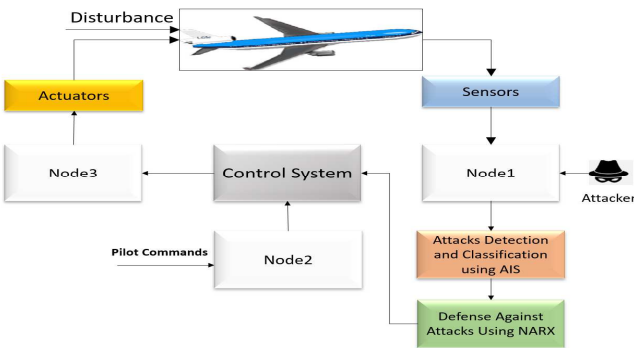


Fig. 8. Simulation Process

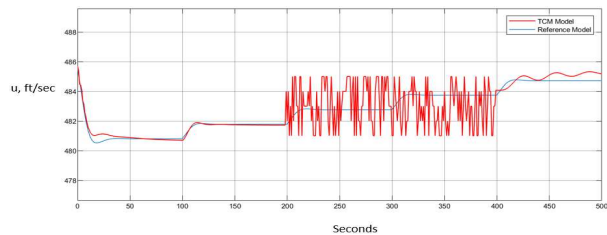


Fig. 9. Aircraft Speed “u, ft/sec” under Attack

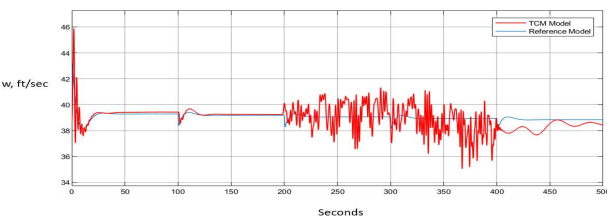


Fig. 10. Aircraft Velocity “w, ft/sec” under Attack

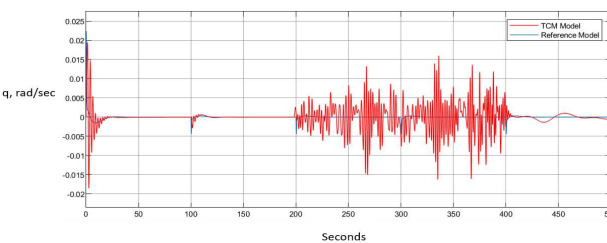


Fig. 11. Aircraft Body Pitch Rate “q, rad/sec” under Attack

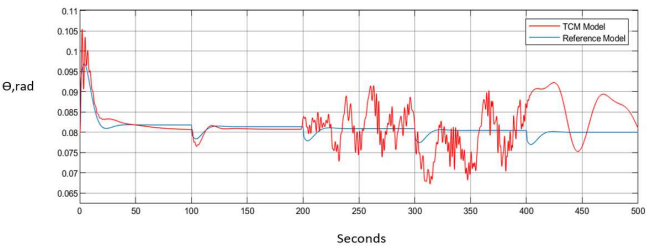


Fig. 12. Aircraft Euler Angle for Pitch “ Θ , rad” under Attack

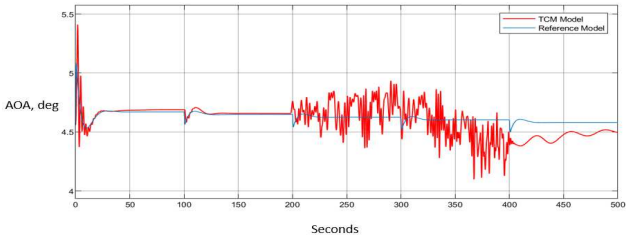


Fig. 13. “AOA, deg” of the Aircraft under Attack

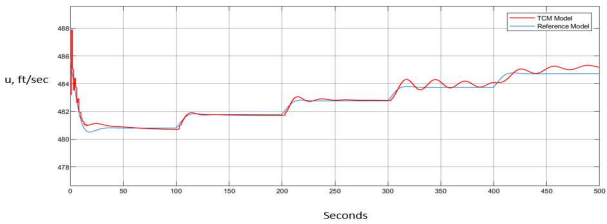


Fig. 14. Aircraft Speed “u, ft/sec” under Attack Using Proposed Methods

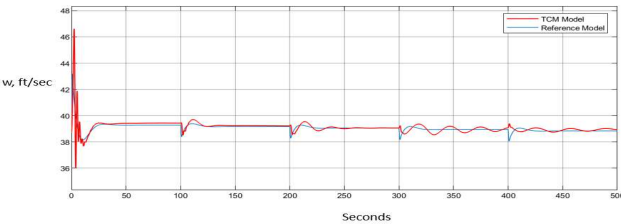


Fig. 15. Aircraft Speed “w, ft/sec” under Attack Using Proposed Methods

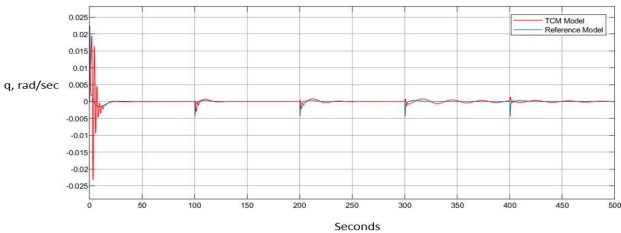


Fig. 16. Aircraft Body Pitch Rate “q, rad/sec” under Attack Using Proposed Methods

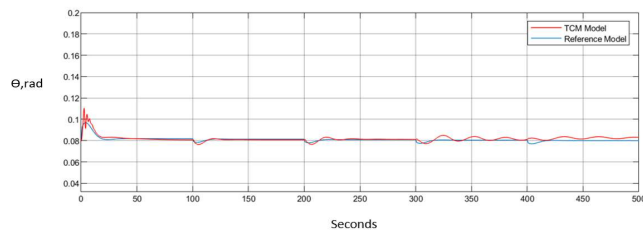


Fig. 17. Aircraft Euler Angle for Pitch “ Θ , rad” under Attack Using Proposed Methods

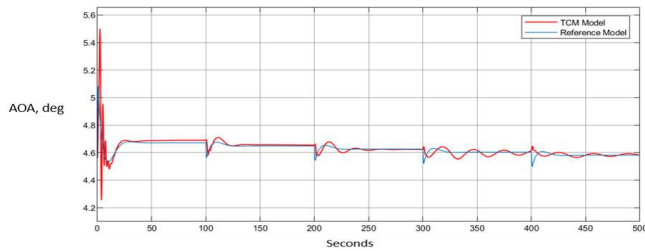


Fig. 18. “AOA, deg” of the Aircraft under Attack Using Proposed Methods

VI. CONCLUSIONS

In conclusion, this research paper proposed a novel and resilient approach to enhance the cyber-security of ACPS systems. First, an aircraft model was transformed into an ACPS by integrating a communication system into the aircraft selected model using the SimEvents toolbox. Second, an effective detection algorithm was developed based on the AIS and was successfully used to detect suspicious communication network packets due to cyber-attacks and dropped the packets out. Third, a NARX neural network was successfully trained to estimate and reconstruct the dropped suspicious packets by the detection algorithm. The developed detection and defense algorithms were tested by the simulation to confirm their validity. The accuracy of the detection algorithm was tested using TPR and TNR measures, and its accuracy was found to be close to 96%. The cyber-security defense algorithm was also tested using the coefficient of determination parameter, and the R-value was found to be higher than 0.99. Therefore, the proposed approach for cyber-security defense can be used with real-time ACPS to mitigate the effect of the sensor spoofing attacks on aircraft, hence enhancing aircraft flights' safety and maintaining performance.

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